



Práctica cero: Mecánica pulmonar

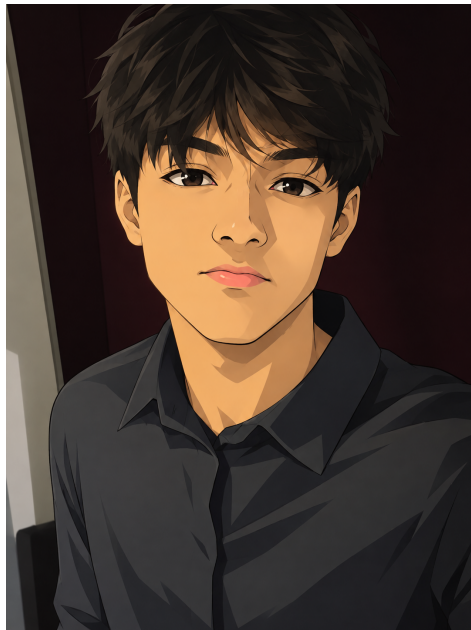
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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system("sistema")
parameters.StopTime = '10'
```

```
parameters =
struct with fields:
    StopTime: '10'
```

```
parameters.Solver = 'ode15s'
```

```
parameters =
struct with fields:
    StopTime: '10'
    Solver: 'ode15s'
```

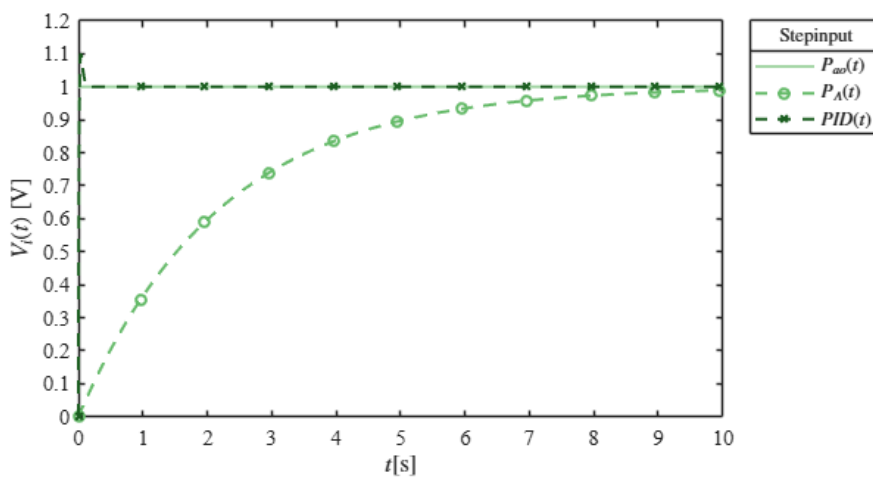
```
parameters.MaxStep = '1e-3'
```

```
parameters =
struct with fields:
    StopTime: '10'
    Solver: 'ode15s'
    MaxStep: '1e-3'
```

```
% set_param('sistema/PA(t): CE/PA','1-D array');
```

Respuesta al escalón

```
set_param('sistema/S1', 'sw', '1')
set_param('sistema/Pao(t)', 'sw', '1')
x1 = sim('sistema', parameters);
plotsignals(x1.t, x1.Pao, x1.PA, x1.PID, 'Step')
```



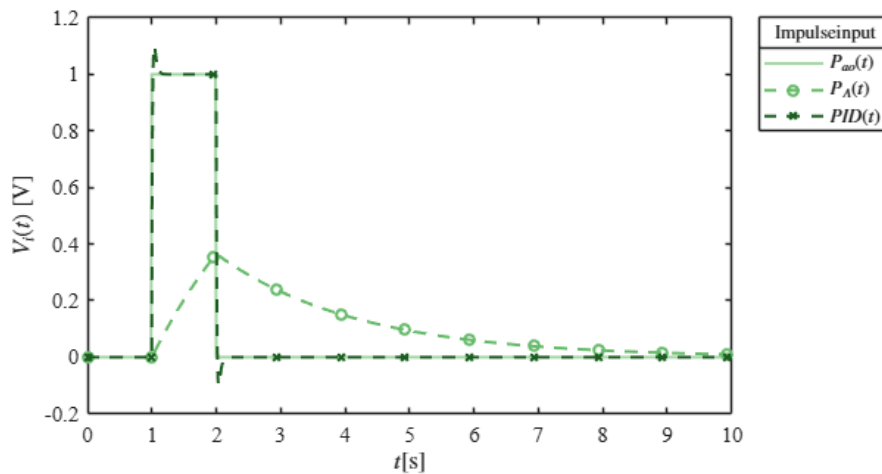
Respuesta al impulso

```
set_param('sistema/S1', 'sw', '0')
```

```

set_param('sistema/Pao(t)', 'sw', '1')
x2 = sim('sistema', parameters);
plotsignals(x2.t, x2.Pao, x2.PA, x2.PID, 'Impulse')

```

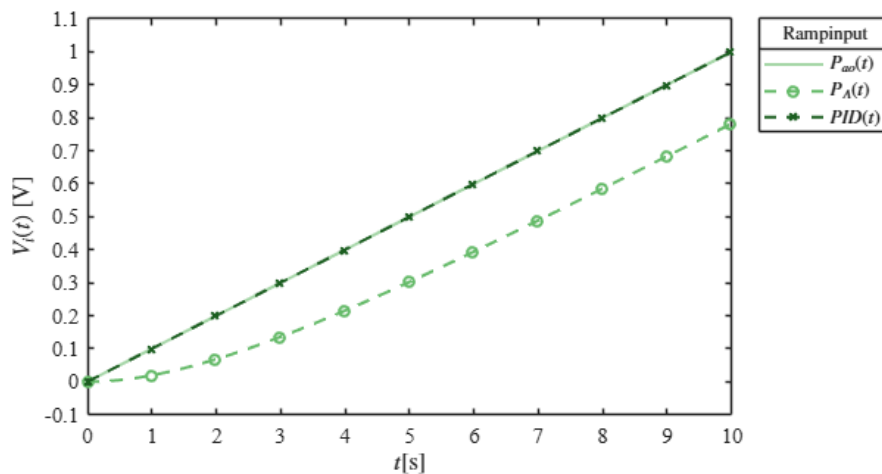


Respuesta a la rampa

```

set_param('sistema/S2', 'sw', '1')
set_param('sistema/Pao(t)', 'sw', '0')
x3 = sim('sistema', parameters);
plotsignals(x3.t, x3.Pao, x3.PA, x3.PID, 'Ramp')

```

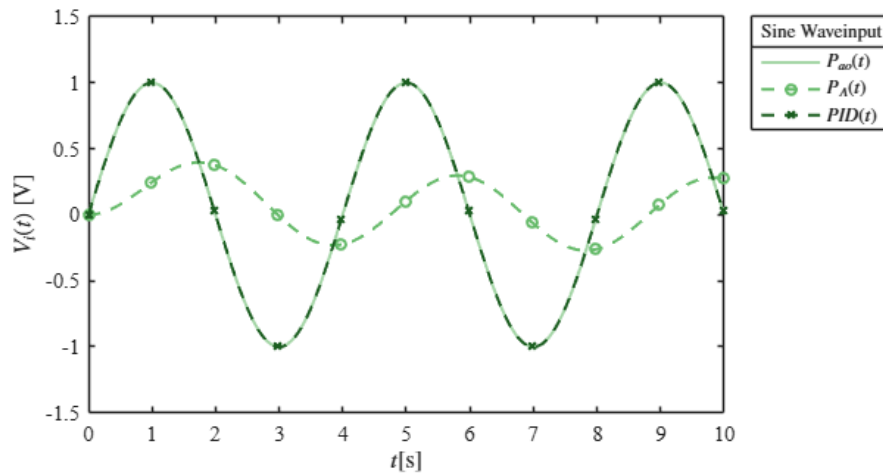


Respuesta a la función sinusoidal

```

set_param('sistema/S2', 'sw', '0')
set_param('sistema/Pao(t)', 'sw', '0')
x4 = sim('sistema', parameters);
plotsignals(x4.t, x4.Pao, x4.PA, x4.PID, 'Sine Wave')

```



Funcion: Respuestas de la simulacion

```
function plotsignals(t,Pao,PA,PID,signal)
    set(gcf,'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [165,214,167]/255;
    clr2 = [102,187,106]/255;
    clr3 = [27,94,32]/255;
    order = [clr1;clr2;clr3];colororder(order)

    plot(t,Pao,'-',t,PA,'--o',t,PID,'--
x','LineWidth',1.5,'MarkerSize',5,'MarkerIndices',1:1e3:length(t));
    L = legend('$P_{ao}(t)$','$P_A(t)$','$PID(t)$');
    set(L,'interpreter','latex','location','northeastoutside','orientation','vert
ical')
    title(L,[signal,'input'])

    xlabel('$t[s]','interpreter','latex')
    ylabel('$V_i(t)$ [V]','interpreter','latex')

    xlim([0 10]); xticks(0:1:10)

    if signal == "Step"
        ylim([0 1.2]); yticks(0:0.1:1.2)
    elseif signal == "Implulse"
        ylim([-0.2 1.2]); yticks(-0.2:0.1:1.2)
    elseif signal == "Ramp"
        ylim([-0.1 1.1]); yticks(-0.1:0.1:1.1)
    elseif signal == "Sinusoidal"
        ylim([-1.2 1.2]); yticks(-1.2:0.2:1.2)
    end
```

```
exportgraphics(gcf,[signal,'_matlab.pdf'],'ContentType','vector')  
end
```